NEWTON SCHOOL

ALGEBRA 1 | ANSWER KEY

Pythagorean Theorem & Distance Formula — Review

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Standard: *CCSS.MATH.CONTENT.8.G.B.7 & 8.G.B.8*

Part A — Find the Missing Side

Problem 1

Legs $a = 6$ and $b = 8$. Find c.

Pythagorean Theorem | 3-4-5 family

$a^2 + b^2 = c^2$

$6^2 + 8^2 = c^2$

$36 + 64 = c^2$

$c^2 = 100$

$c = \sqrt{100} = 10$ (positive root only — length cannot be negative)

$\boxed{c = 10}$

Problem 2

Legs $a = 5$ and $b = 12$. Find c.

Pythagorean Theorem | 5-12-13 triple

$5^2 + 12^2 = c^2$

$25 + 144 = c^2$

$c^2 = 169$

$c = 13$

$\boxed{c = 13}$

Problem 3

One leg $a = 9$, hypotenuse $c = 15$. Find b.

Pythagorean Theorem | Solve for a leg

$a^2 + b^2 = c^2 \;\rightarrow\; b^2 = c^2 - a^2$

$b^2 = 15^2 - 9^2 = 225 - 81 = 144$

$b = \sqrt{144} = 12$

Notice: this is a 3-4-5 triangle scaled by 3 → (9, 12, 15).

$\boxed{b = 12}$

Problem 4

Hypotenuse $c = 25$, one leg $b = 24$. Find a.

Pythagorean Theorem | 7-24-25 triple

$a^2 = c^2 - b^2$

$a^2 = 25^2 - 24^2 = 625 - 576 = 49$

$a = \sqrt{49} = 7$

$\boxed{a = 7}$

Part B — Pythagorean Word Problems

Problem 5

13 ft ladder against a wall, base 5 ft from wall. How high does it reach?

Word problem | Ladder = hypotenuse

The ladder is the hypotenuse ($c = 13$). The base distance is one leg ($a = 5$). The wall height is the unknown leg (b).

$b^2 = 13^2 - 5^2 = 169 - 25 = 144$

$b = 12$ ft

5-12-13 triple again. The ladder itself is the longest side — never a leg.

12 ft

Problem 6

Rectangular stage 9 ft by 12 ft. Find the diagonal.

Rectangle diagonal | 9-12-15 (scaled 3-4-5)

The diagonal of a rectangle is the hypotenuse of a right triangle whose legs are the two sides.

$d^2 = 9^2 + 12^2 = 81 + 144 = 225$

$d = 15$ ft

15 ft

Part C — Distance Formula

Problem 7

Distance between $(1, 2)$ and $(4, 6)$.

Distance Formula | 3-4-5 triple

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{3^2 + 4^2}$

$d = \sqrt{9 + 16} = \sqrt{25} = 5$

$d = 5$

Problem 8

> Distance between $(-3, 4)$ and $(2, -8)$.

Distance Formula | Negative coordinates

$d = \sqrt{(2-(-3))^2 + (-8-4)^2}$

$d = \sqrt{(5)^2 + (-12)^2} = \sqrt{25+144}$

$d = \sqrt{169} = 13$

Subtraction order does not matter once you square — $(-5)^2 = (5)^2$.

$\boxed{d = 13}$

Part D — Mixed Challenge

Problem 9

Stage at $(2, 1)$, exit at $(7, 13)$. Distance in meters.

Distance Formula | Word context

$d = \sqrt{(7-2)^2 + (13-1)^2}$

$d = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169}$

$d = 13$

13 meters

Problem 10

$\triangle ABC$: $A(0, 0)$, $B(8, 0)$, $C(8, 6)$. (a) Find AC by Distance Formula. (b) Verify with Pythagorean Theorem.

Distance Formula | Pythagorean Theorem | Two methods agree

(a) Distance Formula:

$AC = \sqrt{(8-0)^2 + (6-0)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$

(b) Pythagorean Theorem:

Side AB runs from $(0, 0)$ to $(8, 0)$ $\rightarrow$ length $= 8$ (horizontal leg).

Side BC runs from $(8, 0)$ to $(8, 6)$ $\rightarrow$ length $= 6$ (vertical leg).

$AB^2 + BC^2 = AC^2 \;\rightarrow\; 8^2 + 6^2 = 64 + 36 = 100$

$AC = \sqrt{100} = 10$ ✓ matches part (a).

The Distance Formula is literally the Pythagorean Theorem in disguise — the differences $x_2 - x_1$ and $y_2 - y_1$ are the legs of a right triangle.

$AC = 10$